

1) ferretic and ferritic - Martensitic (f/M) steels are being looked at for advanced reactor structural materials because they have advantageous mechanical properties, are relatively cheap, and their microstructure can be altered with the application of heat treatments. 68

Some considerations when optimizing the composition for f/M steels include alloying additions, and which heat treatment will lead way to an optimal microstructure. Also taking into account which phase should be stabilized should also be considered.

- needed detail on alloying additions...

2) ferritic steels swell considerably less than austenitic steels due to several factors including: migration energy of interstitials in f/M steels is larger than in austenite, the barrier energy of vacancies in f/M steels is lower, than in austenite, the interaction of dislocations plays a role, as well as the complex microstructure ferritic steels present which acts as sinks for defects.

3) The ductile to brittle transition temperature (DBTT) is  $0.3 T_m$ , such that  $T_m$  is the melting temperature of the material. - nope...

F/M alloys have an interesting microstructure where martensite laths are seen, coarse  $M_{25}$  particles are present, and  $MC_x$  particles may be present in Cr rich areas.

Under in-reactor conditions these  $MC_x$  particles are known to affect embrittlement. ✓

Also, during irradiation the material is more prone to the build up of defects (interstitials, vacancies, voids) which affects the ductility of the material. (more brittle).

- some of it 7/10

4) FeCrAl alloys are of interest as an advanced cladding material especially in light of the Fukushima accident. FeCrAl alloys form an  $Al_2O_3$  layer which is high T steam resistant, more so than prior used Zircaloy cladding. 6/8

FeCrAl alloys are also being looked into for accident tolerant fuels. ✓

Considerations exist for balancing the amount of Cr, and Al in order for the alloy to be strong enough, yet ductile enough to account for swelling. X - balance corrosion/embrittlement

5) In Oxide Dispersion Strengthened (ODS) steels, oxide particles usually in the form of yttria oxide, or yttria titanium oxide particles act as sites for dislocation pinning. Thus, this makes them favorable in high defect inducing environments such as in a nuclear reactor.

Furthermore, the dispersion of these oxide particles increases strength and gives the steel favorable mechanical properties. These oxide particles also act as defect sinks.  
4/6

6) Ni alloys have high temperature resistance, and advantageous mechanical properties. Many Ni-based superalloys (Haynes) are being looked into for use in molten salt reactors.  
4/8

During the Molten Salt Reactor Experiment at Oak Ridge National Lab, a Ni-based alloy (Alloy N or Hastelloy N) was used, and is still being considered for use in future MSR's to be deployed.

Some disadvantages exist of Ni alloys including limited data, and limited qualifications for use in a nuclear reactor.  
- He transmutation, irradiation hardening...

7) U-Si and U-Al fuel are both metallic fuels being used in research reactors.

U-Si was one of the first fuels to be approved.

U<sub>3</sub>Si<sub>2</sub> was approved in 1988. UAl<sub>4</sub> vs U-Si?

8/12 Research reactor fuels are dense in U. However, the use of reactor fuels (U-Al, U-Si) is not best suited for high temperatures. There exist differences in the swelling behaviors of U<sub>3</sub>Si and U<sub>3</sub>Si<sub>2</sub>. Swelling behavior in U<sub>3</sub>Si<sub>2</sub> is comprised of large bubbles and voids. While swelling occurs in U<sub>3</sub>Si, its extent of swelling is less than in U<sub>3</sub>Si<sub>2</sub>. - 3rd phases inserted, less swelling in U<sub>3</sub>Si<sub>2</sub>

U-Al fuels can form intermetallics UAl<sub>x</sub> such as UAl<sub>2</sub>, UAl<sub>3</sub>, and UAl<sub>4</sub> which are all amorphous and experience swelling - very limited.

U-Si fuels do not form these intermetallic phases but swelling is still experienced.

8) It is favorable for the gamma-UMo phase to be the dominant phase in-reactor due to its properties.

4/7 Various processing and alloying additions can be applied to U-Mo for its gamma phase to be stabilized during in-reactor use. (BCC vs. FCC)  
- not really... , processing somewhat. no FCC

9) The solidus / liquidus gap makes the fabrication of solid metallic U-Mo fuels more difficult.

furthermore, stabilisation of the gamma - UMo phase during fabrication is also difficult. However, U-Mo fuel is better suited for higher temperatures compared with U-Al and U-Si fuels.

- didn't really hit it

10) As the U-Mo fuel experiences higher burnup the extent of fission bubbles increase. At low burnups, not many fission bubbles, and not much porosity is present. However, at higher burnup rates porosity is increased with many more ~~more~~ fission bubbles present.

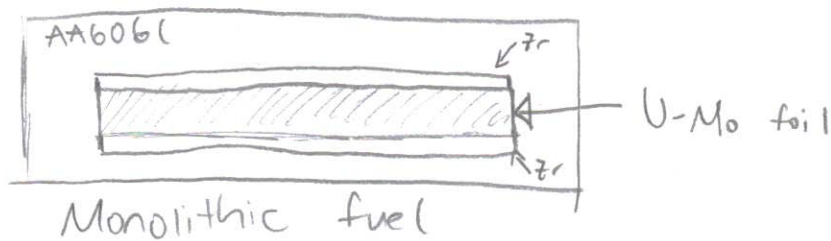
- super-lattice, grain refinement, intergranular bubbles

The amount of fission bubbles, and the buildup of fission bubbles can be problematic if the fuel / cladding assembly does not feature a gap or plenum to take into account this swelling and this change in volume.

- 11) The interaction layer <sup>(lead to fuel failure + Swelling)</sup> is of interest in U-Mo fuels because it gives insights in fuel-cladding interaction, burnup, etc. Monolithic fuel strategies have been deployed to limit the amount of interaction layer in which have cladding (AA6061), and two Zr layers encompassing a U-Mo fuel foil, as seen below.

Monolithic fuel strategy:

-one option



- 12) A research reactor environment is quite different than in a commercial LWR environment. In a research reactor typically doses are lower, and operating temperatures are lower. Furthermore, the fuel in a research reactor is often more Uranium dense. So, Al is ideally suited for a research reactor environment because of the lower doses and temperatures than typically seen in a commercial LWR.
- needed more

Thank You for a great semester! 😊